

Mini ERP + CRM Operations Portal

Official Demo Video Recording Script & Click-by-Click Guide | Candidate: Suraj Mane

GitHub Repository: <https://github.com/manesuraj14/mini-erp-crm> | Target Duration: 5 - 7 Minutes

1. Pre-Recording Setup & Verification

Duration / Focus: 2 Minutes Before Starting

- Start Backend: In terminal run ``cd D:\ERP_Project\backend && npm run dev`` (Port 5000)
- Start Frontend: In terminal run ``cd D:\ERP_Project\frontend && npm run dev`` (Port 5173)
- Browser Tab 1: <http://localhost:5173/login> (Fullscreen window)
- Browser Tab 2: <https://github.com/manesuraj14/mini-erp-crm>
- Recording Tool: Open Loom (loom.com), OBS Studio, or press Win + G on Windows.

Segment 1: Introduction & Tech Stack

Duration / Focus: 0:00 - 0:45

🎯 ON SCREEN ACTION:

Start recording on the Login Screen (<http://localhost:5173/login>). Keep mouse still for 2 seconds.

🗣️ SPOKEN DIALOGUE (READ THIS):

"Hello everyone! My name is Suraj Mane, and today I am excited to present my submission for the Full Stack Developer Case Study: the Mini ERP and CRM Operations Portal. This portal is designed for a wholesale and distribution business to streamline core daily operations across four key departments: Sales, Warehouse, Accounts, and Admin. For the tech stack: On the backend, I built a robust REST API using Node.js, Express, TypeScript, and Prisma ORM connected to a MySQL database. On the frontend, I developed a clean, responsive admin dashboard using React 19, TypeScript, Vite, and Tailwind CSS. Let us dive into the live application."

Segment 2: Authentication & Role-Based Access Control

Duration / Focus: 0:45 - 1:30

🎯 ON SCREEN ACTION:

Hover over the 1-Click Demo Logins box on the login screen. Click the ADMIN demo button to enter. Point cursor to "Active Role: ADMIN" in top navbar, then open the Role Switcher dropdown.

🗣️ SPOKEN DIALOGUE (READ THIS):

"To make evaluation fast and seamless, I implemented both standard JWT authentication and convenient one-click demo login buttons for all four required roles. Let us log in as the Administrator. As you can see, the dashboard gives an instant operational overview: total active clients, real-time stock alert counters, sales challan statuses, and total invoiced revenue. Notice in the top navigation bar, I have included a live Role Switcher. When we switch between roles, the sidebar dynamically filters the modules to enforce strict Role-Based Access Control. Admin has full access, Sales focuses on CRM and Challans, Warehouse manages inventory, and Accounts handles invoicing."

Segment 3: Customer CRM Module

Duration / Focus: 1:30 - 2:45

🔗 ON SCREEN ACTION:

Click Customer CRM in the sidebar. Type "Apollo" in the search box to demonstrate instant debounced search. Click "Add Customer" button, fill in sample details (Apex Healthcare Hub, Wholesale, Active), and submit. Click the Eye icon to open Customer Detail Drawer and add a follow-up note in the timeline.

🗣️ SPOKEN DIALOGUE (READ THIS):

"Now let us explore the Customer CRM Module. The table displays customer contact details, business entity, GSTIN, customer type—such as Retail, Wholesale, or Distributor—and current lead status. We have real-time search across names, phone numbers, and emails, along with multi-parameter filter chips. Let us create a new customer... It is instantly saved into our MySQL database. Next, let us open the Customer Detail Drawer. Here we have a dedicated Follow-up Timeline. Sales team members can log call summaries and meeting outcomes. Each note is saved with the operator name and an immutable timestamp, giving the company a complete interaction history."

Segment 4: Product & Inventory Module

Duration / Focus: 2:45 - 4:00

🔗 ON SCREEN ACTION:

Click Inventory & Stock in sidebar. Point out color badges: Red (Out of Stock), Amber (Low Stock Alert), Green (Healthy). Check "Low Stock Alerts Only" checkbox. Click "Adjust Stock" on Amoxicillin, choose Stock IN, quantity 30, reason "Supplier batch #PO-2026-92", and confirm.

🗣️ SPOKEN DIALOGUE (READ THIS):

"Next is the Product and Inventory Module. Warehouse managers can track current stock levels, warehouse locations, and minimum threshold alert buffers. The system features visual stock badges: Green for healthy inventory, Amber for Low Stock Warnings, and Red for Out of Stock. We can toggle Low Stock Alerts Only to immediately identify SKUs that need replenishment. Let us perform a stock adjustment for Amoxicillin... We will log a Stock IN movement of 30 units with a supplier batch reference. When confirmed, the inventory updates instantly, and an immutable entry is added to our Stock Movement Audit Trail."

Segment 5: Sales Challan Module & Crucial Business Logic

Duration / Focus: 4:00 - 5:30 (HIGHEST EVALUATION WEIGHT)

🔗 ON SCREEN ACTION:

Click Sales Challans. Click "Create Sales Challan". Select Apollo Healthcare. Pick Paracetamol. DEMONSTRATE NEGATIVE STOCK PREVENTION: enter quantity 999 and click Confirm. Show error banner. Change quantity to 10, add second item (Omeprazole 5 units), click Confirm. Show Invoice Modal with snapshot prices, click Print/Export PDF button, then return to Inventory to show stock dropped by 10.

🗣️ SPOKEN DIALOGUE (READ THIS):

"Now let us examine the heart of the system: the Sales Challan Module. Let us click Create Sales Challan. The system automatically generates a sequential order number. First, let us test our core business constraint: Negative Stock Prevention. If a sales representative attempts to order 999 units when only 450 are available... our backend transaction rejects the request with HTTP 400 Bad Request and rolls back, ensuring stock never goes negative. Now let us enter a valid quantity of 10 units and add a second product line item. When I click Confirm and Deduct Stock, the backend executes an ACID transaction: it decrements warehouse stock, logs an OUT stock movement, and confirms the order. Here in the invoice view, notice the Product Snapshot Preservation. The product name, SKU, and unit price are permanently snapshot at the exact time of order. If an administrator later updates catalog prices, this historical invoice remains 100 percent immutable. We can also click Print or Export PDF to generate a delivery challan ready for dispatch. And if we switch to the Inventory page, you can see that the stock was reduced by exactly 10 units in real-time."

Segment 6: Codebase, Docker & Postman Suite

Duration / Focus: 5:30 - 6:30

🎯 ON SCREEN ACTION:

Switch window to VS Code or open GitHub Repository (<https://github.com/manesuraj14/mini-erp-crm>). Briefly show backend/prisma/schema.prisma, docker-compose.yml, postman_collection.json, and .github/workflows/ci.yml.

🗣️ SPOKEN DIALOGUE (READ THIS):

"Behind the scenes, the codebase is cleanly architected: The database is structured in Prisma with full relational integrity and seeded test accounts. Our challan confirmation logic runs in an atomic Prisma transaction to prevent race conditions. For deployment, I have included a docker-compose.yml that boots MySQL, the backend API, and the React frontend in one command. I have also provided a complete Postman v2.1 collection with automated Bearer token passing, a GitHub Actions CI/CD workflow, and comprehensive setup documentation in the README.md."

Segment 7: Wrap-up & Submission Conclusion

Duration / Focus: 6:30 - 7:00

🎯 ON SCREEN ACTION:

Return to browser showing the application Dashboard or GitHub repo homepage.

🗣️ SPOKEN DIALOGUE (READ THIS):

"To summarize: We have implemented all four core modules—Authentication with RBAC, Customer CRM with follow-up history, Product Inventory with low-stock alerts, and Sales Challans with atomic stock reduction and price snapshots—plus bonus containerization and CI/CD. The code is pushed and publicly accessible on my GitHub at github.com/manesuraj14/mini-erp-crm. Thank you very much for reviewing my case study submission!"

🏆 Top 4 Tips for a High-Scoring Video Presentation:

1. Pace: Speak calmly and clearly. Do not rush through the screens.
2. Cursors: Hover over buttons for 1 second before clicking so reviewers can follow your actions.
3. Mistakes: If you make a slip of the tongue, pause for 2 seconds and re-say that line. No need to re-record!
4. Sharing: If using Loom, copy the share link directly into your submission email.